

Heaven's Light is Our Guide



Rajshahi University of Engineering and Technology

Department of Electrical and Computer Engineering

Report 01

Course Title : Analog Electronic Circuits 2- Sessional

Couse Code: ECE-2112

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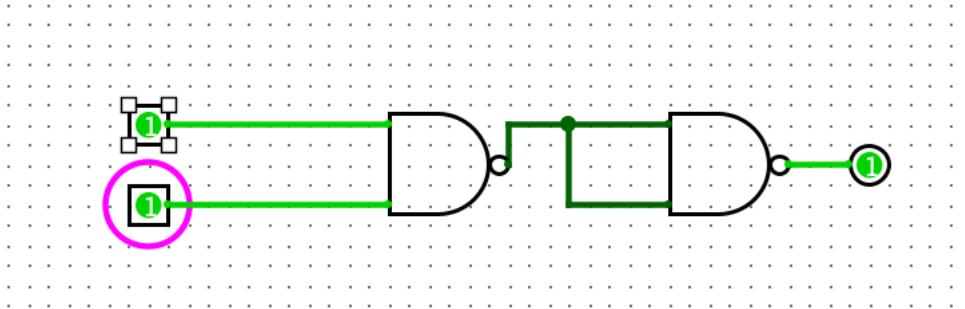
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Experiment No 01:

Design and implement an **AND gate using only NAND gates**. Verify the circuit by comparing the output with the corresponding truth table.

Circuit Diagram:



Truth Table:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

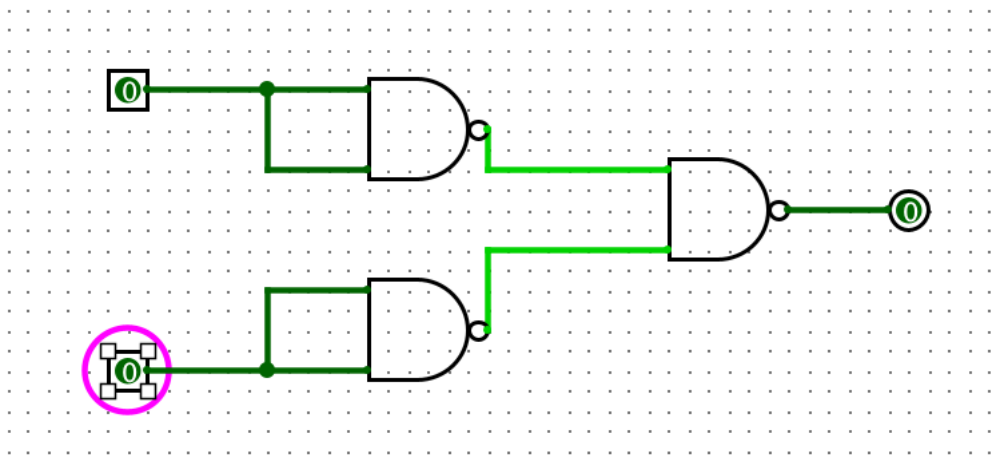
Expression:

$$Y = (A \text{ NAND } B) \text{ NAND } (A \text{ NAND } B)$$

Experiment No 02:

Design and implement an **OR gate using only NAND gates**. Verify the output using the truth table.

Circuit Diagram:



Truth Table:

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

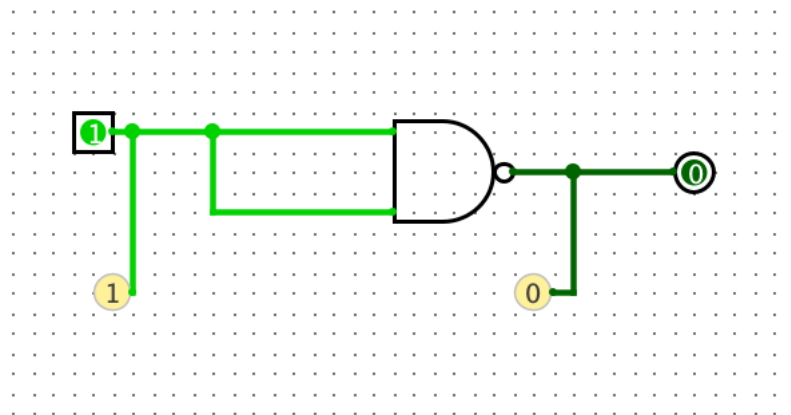
Expression:

$$Y = (A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B)$$

Experiment No 03:

Design and implement a **NOT gate using only NAND gates**. Verify its operation using the truth table.

Circuit Diagram:



Truth Table:

A	Y
0	1
1	0

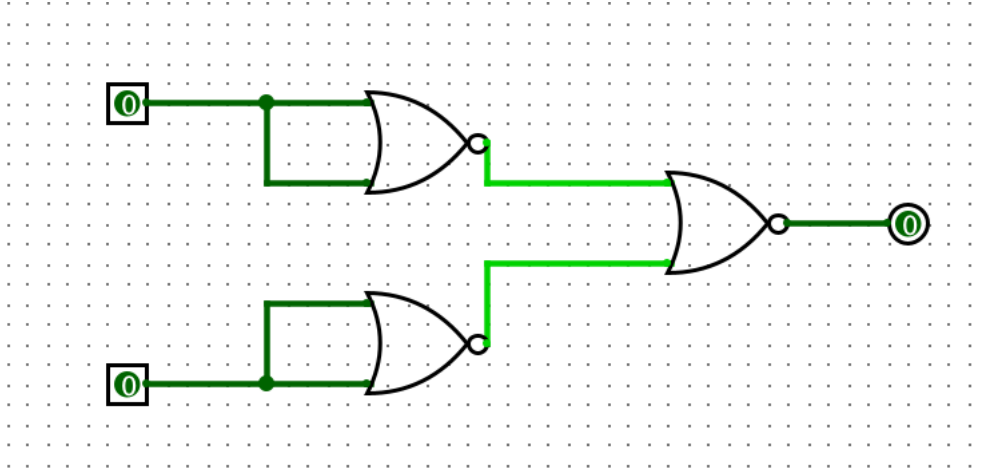
Expression:

$$Y = A \text{ NAND } A$$

Experiment No 04:

Design and implement an **AND gate** using only **NOR gates**. Verify the output using the truth table.

Circuit Diagram:



Truth Table:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

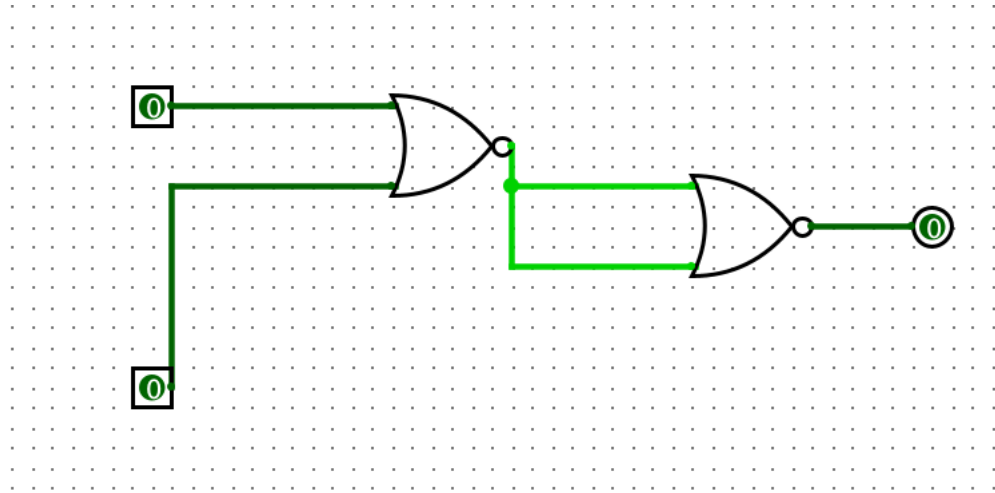
Expression:

$$Y = (A \text{ NOR } A) \text{ NOR } (B \text{ NOR } B)$$

Experiment No 05:

Design and implement an **OR gate using only NOR gates**. Verify the output using the truth table.

Circuit Diagram:



Truth Table:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Expression:

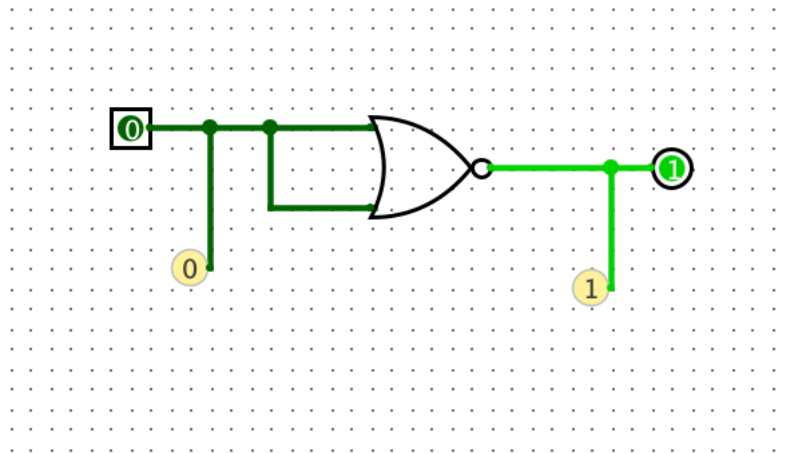
$$Y = (A \text{ NOR } B) \text{ NOR } (A \text{ NOR } B)$$

Experiment No

06:

Design and implement a **NOT gate using only NOR gates**. Verify the output using the truth table.

Circuit Diagram:



Truth Table:

A	Y
0	1
1	0

Expression:

$$Y = A \text{ NOR } A$$

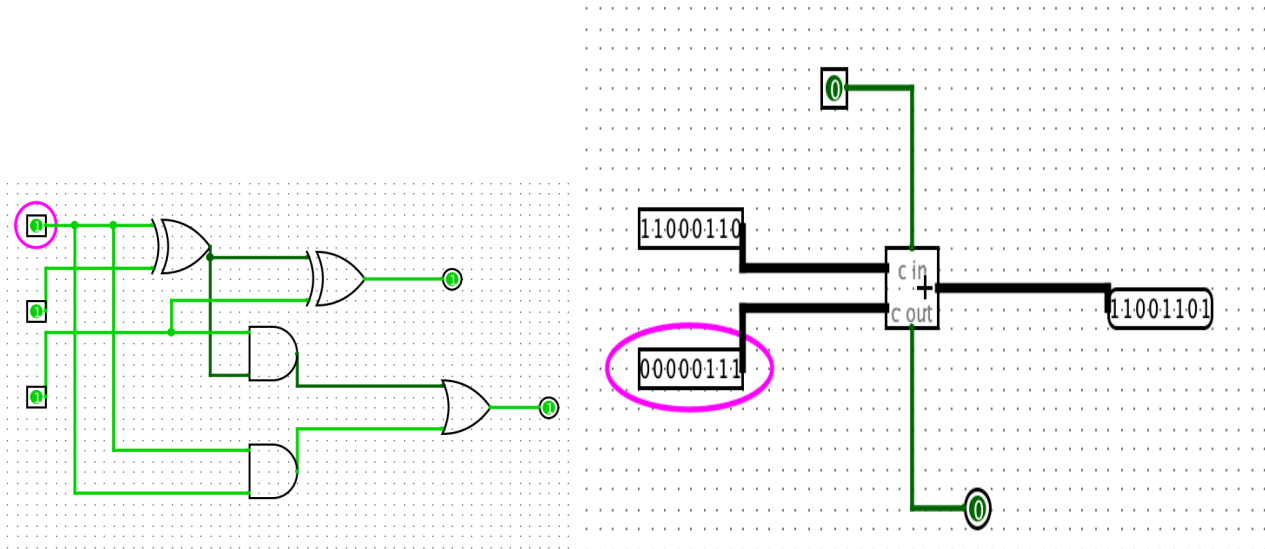
07:

Experiment No

07:

Design and implement a **1-bit Full Adder using logic gates**. Verify the correctness of the circuit by comparing the generated outputs with the theoretical truth table.

Circuit Diagram:



Truth Table:

A	B	C _{in}	Sum	Carry
0	0	0	0	0
0	1	1	1	0
0	0	0	1	0
0	1	1	0	1
1	0	0	1	0
1	1	1	0	1
1	0	0	0	1
1	1	1	1	1

Expression:

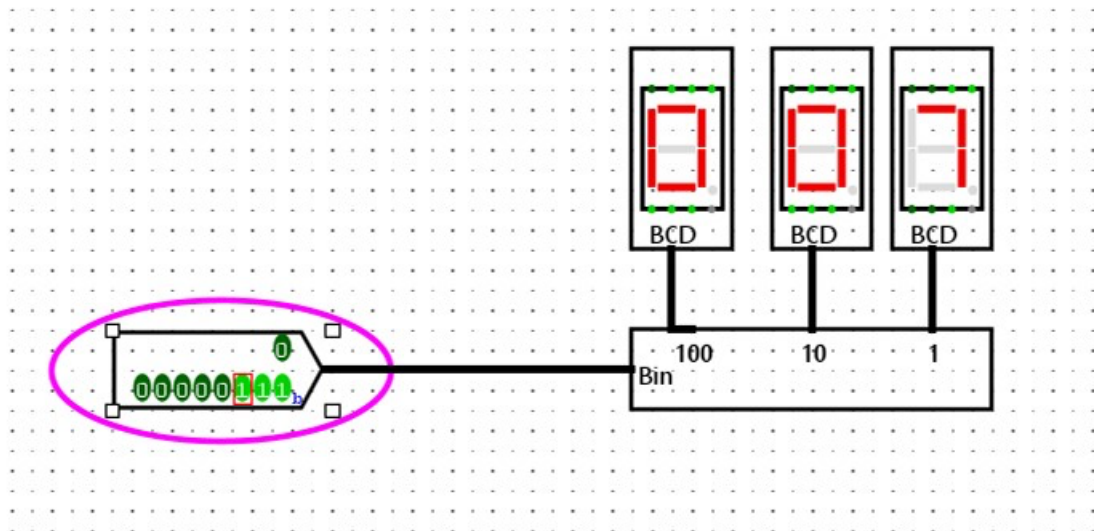
$$Y = A \oplus B \oplus C_{in}$$

Experiment No

08:

Design and implement a **Binary-to-BCD Converter**. Verify the circuit by applying different binary input combinations and comparing the outputs with the expected BCD values.

Circuit Diagram:



Truth Table:

Binary Input	Decimal	Hundreds	Tens	Ones
00000000	0	0	0	0
00000001	1	0	0	1
00000010	2	0	0	2
00000011	3	0	0	3
00000100	4	0	0	4
00000101	5	0	0	5
00000110	6	0	0	6
00000111	7	0	0	7
00001000	8	0	0	8
00001001	9	0	0	9
00001010	10	0	0	0
00001011	11	0	0	1
.....				
1100100	100	1	0	0
1111111	127	1	2	7

Experiment No

Expression: Binary $\rightarrow$ BCD